W/kg on a Climb is a Distribution, not a Verdict: Estimating Cycling Climbing Power with Honest Uncertainty

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Abstract

Every Grand Tour mountain stage now ends the same way: a self-appointed "watts watcher" posts a single relative-power figure — "6.4 W/kg" — and declares the rider a mutant. The point estimate is the problem. The inputs that feed it (rider mass, frontal area, rolling resistance, wind, air density, the timing pulled from television) are not known precisely, so a number quoted to two decimals is false precision dressed as evidence. We describe a small, reproducible estimator that keeps the physics honest about what it does not know. A validated power-balance model (Martin et al., 1998) converts public climb data (length, elevation gain, time) into mechanical power, and a Monte Carlo layer propagates the uncertainty in every uncertain input into a *distribution* of W/kg rather than a single value. The output is a median, a 90% credible interval, and the probability of exceeding a chosen physiological threshold. On the demonstration climb (Alpe d'Huez, 40:00) the estimate is 5.77 W/kg (90% CI 5.38 to 6.33), $P(>6.2) = 9.5\,\%$. Applied to two iconic Pyrenean efforts, the method places Froome (Ax 3 Domaines, 2013) at a median 6.35 W/kg (5.86 to 7.05, $P(>6.2) = 67\,\%$) — straddling the informal "suspicion" line, genuinely ambiguous — and Pogačar (Plateau de Beille, 2024) at 7.00 W/kg (6.46 to 7.77, $P(>6.2) = 100\,\%$) — unambiguously above the old ceiling. The method is deliberately one of *estimation, not accusation*: the model is sound, the inputs are uncertain, and so the result is reported as a range. "Above the old ceiling" is not "doping."

1 Introduction

Relative power output — watts per kilogram of body mass — has become the de facto lie detector of professional cycling. On a sustained climb, where aerodynamic drag is small and gravity dominates, W/kg is closely tied to ascent rate, so a fast climb invites an immediate back-of-envelope estimate and, with it, a verdict. The cultural ritual is familiar: a stage finishes, an estimate circulates, and a figure crosses some remembered "ceiling" for clean performance, at which point suspicion is treated as established fact.

There are two distinct problems with this ritual, and only one of them is about doping. The first is that a single point estimate misrepresents the evidence. The quantities required to turn a climb into a power figure are not measured — they are guessed, and some are guessed badly. Published rider weights are unreliable and mass is the single largest lever on the estimate; wind and drafting are usually unknown; the climb's exact length, gradient and the rider's finishing time are read off television graphics or third-party segments. A figure quoted as "6.4 W/kg" hides all of this behind a decimal point it has not earned.

The second problem is the standard used to interpret the figure. The most widely circulated shortcut is the VAM rule (vertical ascent metres per hour divided by a gradient-dependent factor), popularised as the "Ferrari factor." It is a coaching heuristic, not a peer-reviewed model, and it was never intended to adjudicate guilt. The rigorous alternative has existed since 1998: Martin et al. (1998) derived and experimentally validated a mathematical model of road-cycling power, finding modelled and measured power highly correlated ($R^2 = 0.97$) and not statistically different, with a standard error of estimate of only 2.7 W. That model — gravity, rolling resistance, aerodynamic drag and drivetrain losses — is the defensible backbone for any W/kg estimate.

This paper takes the position that the physics is the easy part and the honesty is the hard part. We pair the validated power-balance model with explicit uncertainty propagation, so that the output is never a single number but a distribution: a median, a 90% credible interval, and an exceedance probability $P(\mathrm{W/kg} > \tau)$ for a user-chosen threshold τ. We situate the result in the power–duration framework of the Record Power Profile (Pinot and Grappe, 2011), which treats sustainable power as a function of effort duration rather than a single scalar. The aim throughout is to estimate a mechanical effort *with* its uncertainty, and to refuse the leap from "above the old ceiling" to "cheating."

2 Methods

2.1 The physical power–balance model

For a rider of total system mass m (rider plus bicycle and kit) ascending a road of gradient θ at ground speed v, the steady-state mechanical power delivered at the rider is the sum of the power to lift the system against gravity, to overcome rolling resistance, and to push air aside, divided by drivetrain efficiency η:

$$P = \frac{1}{\eta}\Big[\underbrace{m\,g\,\sin\theta\,v}_{\text{gravity}} + \underbrace{C_{rr}\,m\,g\,\cos\theta\,v}_{\text{rolling}} + \underbrace{\tfrac{1}{2}\,\rho\,C_dA\,(v+v_w)^2\,v}_{\text{aerodynamic}}\Big], \tag{1}$$

where $g = 9.81\,\mathrm{m\,s^{-2}}$, ρ is air density, C_dA the effective drag area, C_{rr} the rolling-resistance coefficient and v_w a headwind component ($v_w > 0$ is a headwind). Equation (1) is the form validated by Martin et al. (1998); on a steep climb the gravity term dominates, but the rolling and aerodynamic terms remain non-negligible and are retained. Relative power is P/m_{rider}, reported per kilogram of rider body mass.

2.2 Three estimators, from crude to defensible

The implementation exposes three estimators that share the same public inputs — climb length L, elevation gain Δh, time t and an assumed rider mass — and differ in rigour.

(i) Ferrari / VAM heuristic. The social-media shortcut. With $\mathrm{VAM} = \Delta h \cdot 3600/t$ (vertical metres per hour) and a gradient s in percent,

$$\mathrm{W/kg}_{\text{Ferrari}} = \frac{\mathrm{VAM}}{(2 + s/10)\cdot 100}. \tag{2}$$

It is fast, requires no assumptions about mass or drag, and is explicitly labelled an approximation: a coaching rule of thumb, *not* a peer-reviewed estimator. We report it only to show how close — or how far — the crude heuristic lands relative to the physical model.

(ii) Physical point estimate. Equation (1) evaluated once at central values of every parameter ($m_{\text{bike}} = 8.0\,\mathrm{kg}$, $\rho = 1.10\,\mathrm{kg\,m^{-3}}$, $C_dA = 0.32\,\mathrm{m^2}$, $C_{rr} = 0.004$, $v_w = 0$, $\eta = 0.976$), with road speed $v = L/t$ and gradient inferred from L and Δh. This is the best single number the physics can give — and precisely the number we argue should never be reported alone.

Table 1: Monte Carlo priors. Each uncertain input is drawn from a normal distribution; a single `unc_scale` multiplier widens or narrows every standard deviation together (defaults shown). The mass terms carry by far the largest standard deviation relative to their effect on the estimate, which is why mass is the dominant lever.

Input	Symbol	Mean	Std. dev. (default)
Rider mass	m_{rider}	published value	2.0 kg
Bike + kit mass	m_{bike}	8.0 kg	0.5 kg
Air density	ρ	$1.10\,\mathrm{kg\,m^{-3}}$	0.04
Drag area	C_dA	$0.32\,\mathrm{m^2}$	0.03
Rolling coeff.	C_{rr}	0.004	0.0006
Wind component	v_w	$0.0\,\mathrm{m\,s^{-1}}$	$1.5\,\mathrm{m\,s^{-1}}$
Drivetrain eff.	η	0.976	0.005
Elevation gain	Δh	measured	2% of Δh (DEM/GPS)
Climb length	L	measured	1% of L

(iii) Monte Carlo distribution. The estimator we advocate. Rather than evaluating Equation (1) at fixed inputs, we draw $n = 200\,000$ samples for every uncertain quantity from independent priors (Table 1), evaluate the model for each draw, and report the resulting distribution of W/kg by its median, its 5th and 95th percentiles (a 90% credible interval), and the exceedance probability $P(\text{W/kg} > \tau)$.

2.3 Uncertainty propagation and priors

The priors in Table 1 encode where the ignorance actually lives. Rider mass is given a 2 kg standard deviation because published racing weights are systematically unreliable, and because mass enters Equation (1) both in the numerator (the system being lifted) and in the denominator (the body mass we divide by); it is the single largest source of spread in the output. Wind is given a $1.5\,\mathrm{m\,s^{-1}}$ standard deviation centred on zero, reflecting that the true head/tail component on a televised climb is essentially unknown and matters most on shallower gradients. Air density, drag area, rolling resistance and drivetrain efficiency carry narrower, physically motivated priors. Geometry (length and gain) is given small multiplicative noise to represent DEM and GPS error. The priors are deliberately wide and editable in one line of code; a single `unc_scale` multiplier scales every standard deviation at once, so a reader who believes the inputs are better (or worse) constrained can widen or narrow the whole interval coherently.

2.4 Power–duration context

A W/kg figure is meaningless without a duration attached: 7 W/kg for one minute and for forty minutes describe entirely different physiological feats. We therefore interpret each estimate within the Record Power Profile framework of Pinot and Grappe (2011), in which sustainable power follows an approximately hyperbolic relationship with effort duration and the profile across durations acts as a "signature" of a rider's physical capacity. Every estimate in this paper is a sustained climbing effort of roughly 20 to 40 minutes, and should be read as a point on that duration curve, not as a context-free scalar.

3 Results

All numbers below are produced directly by the accompanying code (`wkg_estimator.py` and `compare_climbs.py`); the figures are the unmodified outputs of those scripts.

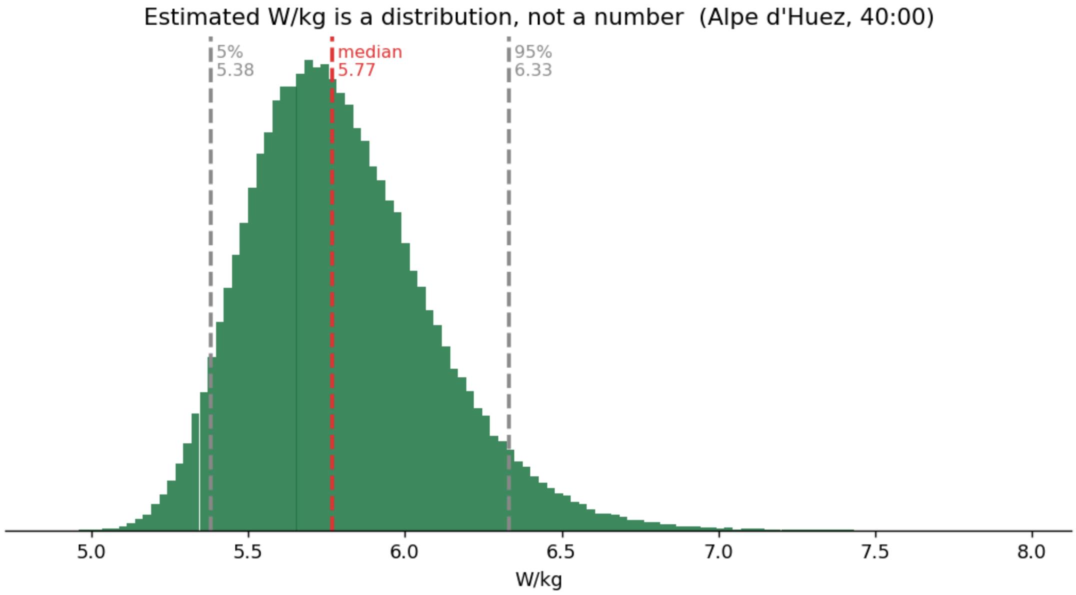

Figure 1: The estimate is a distribution, not a number. Monte Carlo output for the Alpe d'Huez demonstration: median 5.77 W/kg with a 90% credible interval of 5.38 to 6.33 W/kg. The point estimate (5.76 W/kg) locates the centre; the spread is the honest part.

3.1 Demonstration: Alpe d'Huez

The demonstration applies all three estimators to a canonical effort: Alpe d'Huez, 13.2 km at 8.1 %, climbed in 40:00 by a 64 kg rider (VAM $1606\,\mathrm{m\,h^{-1}}$). The Ferrari heuristic returns 5.71 W/kg and the physical point estimate 5.76 W/kg — reassuringly close, confirming the crude rule lands near the validated model for a typical climbing gradient. The Monte Carlo layer turns this into a distribution (Figure 1): median 5.77 W/kg, 90% credible interval 5.38 to 6.33, and $P(\mathrm{W/kg} > 6.2) = 9.5\,\%$. The single number and the distribution agree on the centre; only the distribution also tells you that, given honest input uncertainty, this effort sits comfortably below the 6.2 line about nine times out of ten.

3.2 Two Pyrenean efforts, eleven years apart

The substantive comparison runs the same pipeline on two famous climbs separated by more than a decade (Table 2, Figure 2). Both were widely cited at the time as "too fast to be clean," and the method lets us ask whether that label is supported by the evidence *once the uncertainty is made explicit.*

The two cases land very differently. Froome's 2013 Ax 3 Domaines effort estimates to a median of 6.35 W/kg with a 90% interval of 5.86 to 7.05 and $P(\mathrm{W/kg} > 6.2) = 67\,\%$. The distribution *straddles* the 6.2 line: roughly a third of the credible interval lies below it. This is the quantitative shape of a genuinely ambiguous case — which is precisely why the effort was argued about, inconclusively, for years. A point estimate of 6.35 reads like an indictment; the interval reveals that the evidence does not actually clear the line with confidence.

Pogačar's 2024 Plateau de Beille effort is different in kind. The median is 7.00 W/kg, the 90% interval is 6.46 to 7.77, and $P(\mathrm{W/kg} > 6.2) = 100\,\%$: *every* sample in the credible interval clears the suspicion line, and does so over a climb more than twice as long. Here the uncertainty does not rescue ambiguity — the entire distribution sits above the old ceiling. The same method, applied to two efforts eleven years apart, separates "on the line" from "clearly past it" without ever collapsing either into a single accusatory figure.

Table 2: Estimator outputs for the two climbs, as printed by `compare_climbs.py`. The Monte Carlo column is the result we advocate reporting; the single-figure columns are shown only for comparison. The suspicion threshold is $\tau = 6.2$ W/kg.

	Froome Ax 3 Domaines (2013)	**Pogačar** Plateau de Beille (2024)
Climb length	8.9 km	15.8 km
Gradient	7.48 %	7.93 %
Time	23:14	39:43
VAM	$1715\,\mathrm{m\,h^{-1}}$	$1887\,\mathrm{m\,h^{-1}}$
Assumed mass	67 kg	66 kg
Ferrari (heuristic)	6.24	6.76
Physical (point)	6.35	7.00
MC median	**6.35**	**7.00**
90% CI	5.86 to 7.05	6.46 to 7.77
$P(>6.2)$	67 %	100 %

We stress what this does and does not say. It says that, conditional on the public data and the priors in Table 1, Pogačar's mechanical W/kg on that climb almost certainly exceeded 6.2 and Froome's plausibly did not. It does not say either rider was or was not doping. The 6.2 line is a remembered heuristic ceiling, not a biological constant, and the ceiling itself can move (Section 4).

4 Discussion and Limitations

The estimator is honest about its own scope, and the limitations are not footnotes — they are the point.

4.1 Why the two estimates are not comparable

The model estimates the mechanical power of one effort, and nothing more. It does not say whether a given W/kg is physiologically remarkable, and — more to the point here — it does not place two efforts on a common axis. Reading Froome's 6.35 and Pogačar's 7.00 as two measurements of the same quantity is a category error, for two independent reasons: the efforts differ in duration, and they were ridden in different race contexts. Both differences block a like-for-like reading, and they cut in both directions rather than toward a verdict.

Duration: the clean axis. The least disputable difference is simply how long each power was held: Froome's effort lasted about 23 minutes, Pogačar's nearly 40. By the power–duration relationship of Pinot and Grappe (2011) — the same Record Power Profile framework introduced in Section 2.4 — sustainable power declines as duration grows, so a higher W/kg held for almost twice as long is a strictly harder point on the curve, independent of any assumption about the rider, the weather or the race. This is enough on its own: 7.00 W/kg for 40 minutes and 6.35 W/kg for 23 minutes are not "0.65 apart" in any physiologically meaningful sense, because they sit at different places on the duration curve. The raw subtraction that fuels social-media comparisons is the error this axis exposes.

Race context: a filter that cuts both ways. The second difference is the race situation, and here the literature counsels caution, not a conclusion. Climbing power is conditioned by accumulated load (Leo et al., 2022), by gradient (Valenzuela et al., 2022) and by stage type

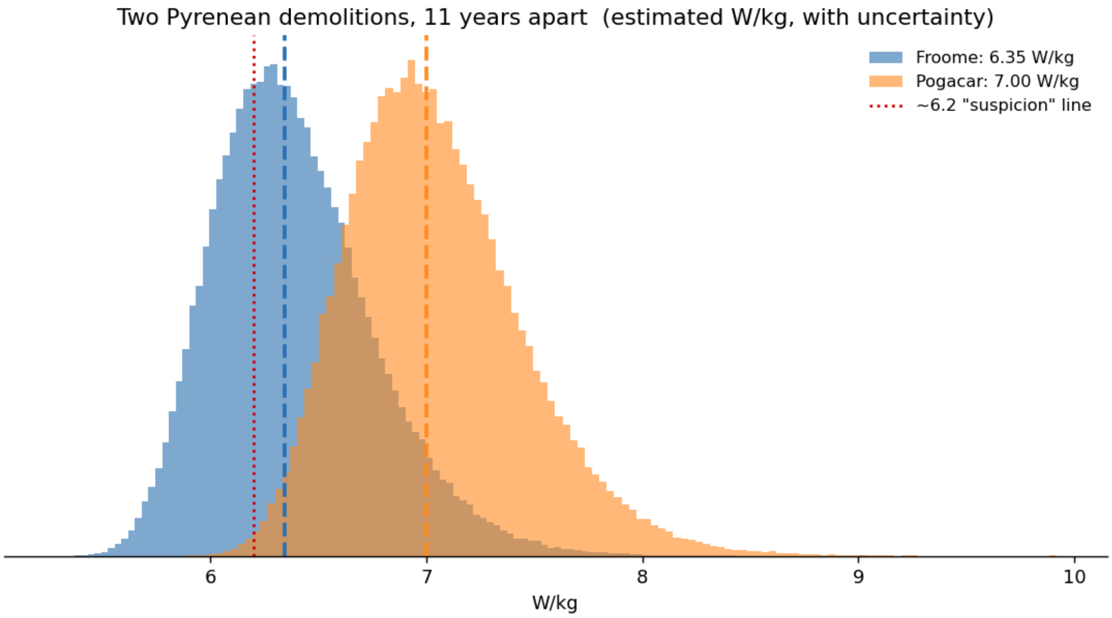

Figure 2: Two Pyrenean efforts, eleven years apart, with their full estimated W/kg distributions. Froome's distribution (median 6.35 W/kg) straddles the informal 6.2 suspicion line (dotted); Pogačar's (median 7.00 W/kg) sits entirely to its right. The overlap on the left tells the whole story: one effort is ambiguous, the other is not.

(Sanders and van Erp, 2021), so the surrounding conditions change what a number means — in either direction. The two efforts sat very differently on these axes. Froome's came in the opening week, on low cumulative race load; but within that stage it followed an earlier hors-catégorie climb (the Col de Pailhères) and was decided by roughly 5 km alone at full effort, so it was neither a fully fresh nor a steady-state test. Pogačar's came at the end of the second week, closing a long mountain stage of roughly 5000 m of climbing in heat. These facts — drawn from the public race record and kept separate from the model outputs — do not establish that either effort was the greater feat; they establish that the two were produced under conditions different enough that the medians should not be stacked side by side. The honest use of context is to block the false comparison, not to extract a new verdict from it.

Read together, the two cases make the same point about a *pair* of estimates that the rest of the paper makes about a *single* one: a W/kg figure is not self-interpreting, and two figures are no more comparable than one is conclusive. Different durations and different race contexts mean the numbers are not on one scale — which is a reason to resist the comparison, in both directions, not a lever for deciding whose climb was more extraordinary or whose was more suspect.

4.2 Scope and limitations

There is no fatigue layer. Most importantly, the model has no representation of accumulated fatigue, heat, nutrition or stage context. It estimates mechanical power, full stop — it does not judge whether that power is "clean." This is a deliberate boundary: adding a physiological-plausibility layer would re-introduce exactly the accusatory leap the method is built to avoid. "Above the old ceiling" has many innocent explanations (altitude acclimatisation, heat protocols, nutrition, aerodynamic and training advances, a genuinely exceptional athlete), and the model cannot distinguish among them.

Measuring performance is intrinsically noisy. Even the underlying quantity is stochastic. Passfield et al. (2017) emphasise that field power data have an inherently variable, stochastic nature that simple summary figures cannot represent — which is the empirical justification for reporting a distribution rather than a point. Our inputs are worse than a power meter on the bike: mass is assumed, wind is unknown, and the geometry and timing are read off television. The credible interval is wide because the honest answer is wide.

Validation is by plausibility, not case-by-case ground truth. We do not have the riders' true power files, so we cannot validate any single estimate against measured power. What anchors the method to reality is published longitudinal data such as Pinot and Grappe (2015), a six-year monitoring case study of a twice top-ten Grand Tour finisher whose recorded power outputs across durations from five minutes to four hours bound the physiological range these estimates should fall within. Our medians for elite climbs are consistent with that range, which is a plausibility check — not a per-case proof.

Scope statement. In one sentence: this is a mechanical estimate of a single effort, built on a validated power model but fed by uncertain inputs (mass, wind, television data), with no fatigue model and no case-by-case empirical validation. Reported as a range, it is informative; collapsed to a point and read as a verdict, it is exactly the false precision it was written to replace.

5 Conclusion

The watts-watcher ritual fails not because the physics is wrong but because a point estimate pretends to a certainty the inputs cannot support. Coupling the validated power-balance model of Martin et al. (1998) with explicit Monte Carlo propagation replaces the single "mutant" figure with a median, a 90% credible interval, and an exceedance probability. On our two test cases that distinction is decisive: Froome's 2013 climb straddles the suspicion line and is genuinely ambiguous ($P(>6.2) = 67\,\%$), while Pogačar's 2024 climb sits entirely above it ($P(>6.2) = 100\,\%$). The method estimates with uncertainty; it does not accuse. The number worth reporting was never a number — it was a range.

Reproducibility. All results, including both figures and every value in Tables 1–2, are produced by the accompanying code (`wkg_estimator.py`, `compare_climbs.py`) and verified by a sanity-test suite (`python -m pytest`).

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